TECHNICAL UNIVERSITY OF MUNICH

School of Computation, Information and Technology
Informatics

Master's Thesis in Informatics

DRAFT - Thesis Proposal: Something With Usability in PIX:E

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Thesis Proposal: Irgendetwas mit Usability in PIX:E

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Disclaimer

I confirm that this Master's thesis is my own work and I have documented all sources and material used.

(Date) (Rai Trouvain)

Acknowledgements

The acknowledgements

Abstract

English Abstract

Zusammenfassung

German Abstract

Contents

1 Introduction

1.1 Context/Motivation

Usability is a crucial, yet often underprioritized aspect of software development. There are different factors that contribute to usability, such as users' understanding of a software's capabilities, or how easy it is to utilize those capabilities. In short, good usability of a software means that it can be used effectively and easily. Usability belongs to the broader subject of user experience (UX), which for instance also includes how engaging a software is for users. The topic of usability is also related to accessibility, which focuses more on whether software can be used by people regardless of factors like age or ability.

1.2 Thesis Goal/Research Questions

The main focus of my thesis are going to be the research and implementation of maintainable, effective, and research-backed principles of usability and accessibility. This includes both established guidelines and state-of-the-art findings.

The software this thesis will work on is pix:e, a toolkit for video game developers (Geheeb, Dyrda, and Geheeb 2024). Specific aspects of this software to consider are the open-source context of the software, the NLP-based features, and the technical background of the intended user base. The aim will be to answer the following questions:

RQ1 How can different approaches/methods to evaluating and implementing usability be combined?

RQ2 How impactful are different usability features in the context of the software domain?

The first question will be addressed by a literature review and usability audit, which includes implementation of usability features. For the second question, the aforementioned improvements will be evaluated in a user study.

2 Background

2.1 Usability Guidelines

A common standard for good usability are Nielsen's heuristics (Nielsen 2024). Another established standard are the Web Content Accessibility Guidelines (WCAG) ("Web Content Accessibility Guidelines (WCAG) 2.1," n.d.), which are an international standard that is explicitly referenced in EU norm 301 549 (TIDO: cite).

(Amershi et al. 2019) proposes guidelines specifically for human-AI interaction.

2.2 Usability In Software Projects

(Lage et al. 2019) did a multi-case study about usability technical debt in software projects, outlining the context and nature of usability aspects that typically contribute to this debt. As for the process of improving usability in softwares, (Llerena et al. 2025) offer a review of evaluation techniques and proposals on how to adapt them to an open-source software (OSS) context.

2.3 Usability User Studies

- System Usability Scale
- TODO

3 Methodology

3.1 Initial Research

The initial phase of research focuses on fundamental guidelines and principles of usability. It also aims to gather any research about usability in the specific domains pix:e falls into, including open source software, planning tools, and design toolkits. Another interesting aspect would be usability of AI/NLP-based tools, as pix:e also includes NLP features.

3.2 Usability Inspection/Audit

Based on the initial research, the next step would be to examine the existing software components (potentially with special focus on some more crucial/developed parts of it). The goal here is to evaluate the current usability and gain an understanding of the scope and context of possible improvements and usability features.

3.3 Improvements - Iteration 1

The first iteration of improvements includes features identified in the previous step, selected considering both maintainability and impact.

3.4 Usability Test (User Study)

The aim of this user study will be to evaluate the impact of the implemented usability improvements. Depending on the available resources and variety of improvements, this will either involve a comparison of the previous version of the software to the updated one, or a comparison of the different new features. For the former setup, a mainly quantitative study with two groups of participants might be suitable, whereas for the latter, a qualitative study may be the more appropriate option.

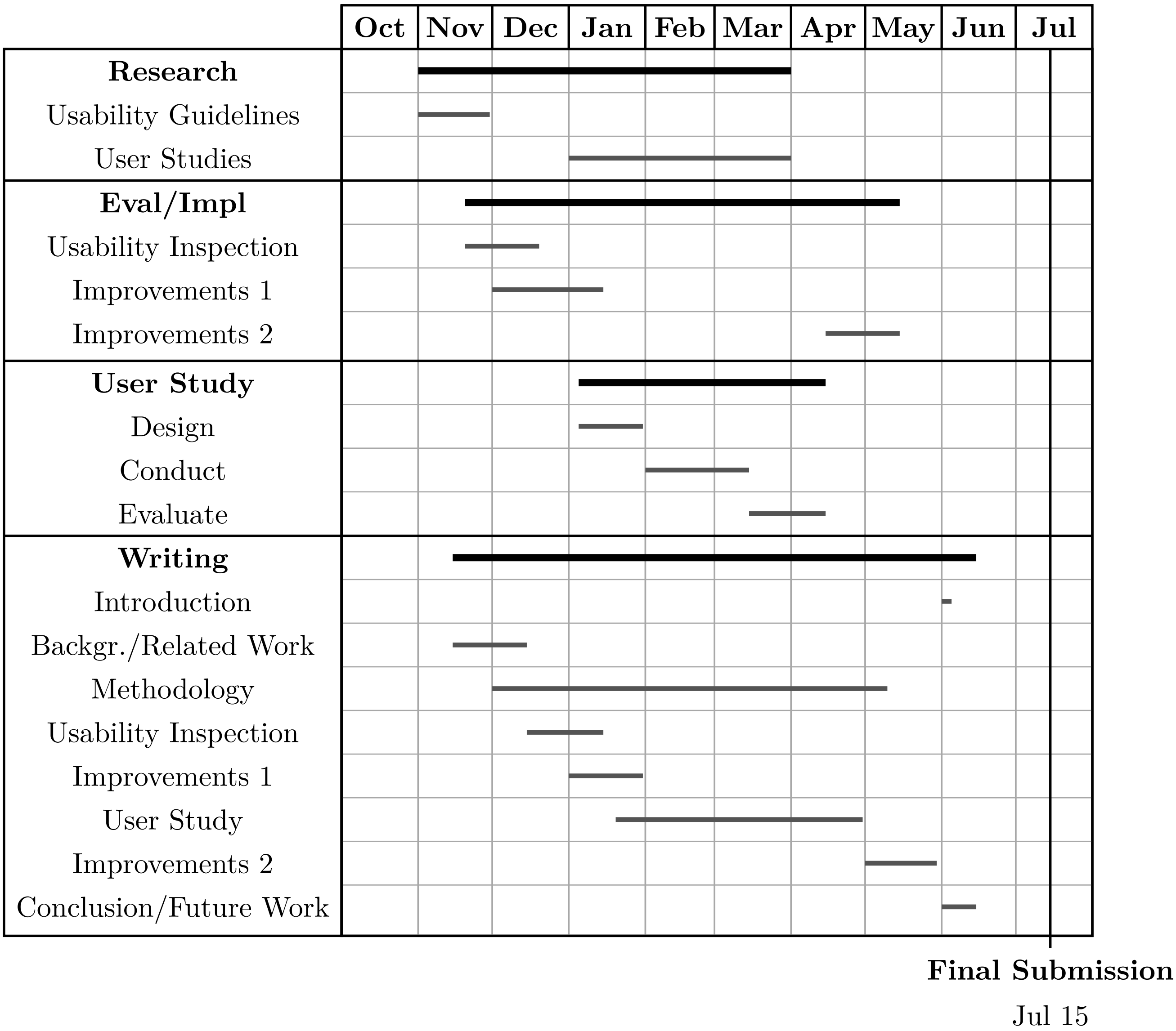
The two central questions for this study are:

1. Did the implemented improvements actually improve usability?
2. What other usability features might be interesting/desired by users?

3.5 Improvements - Iteration 2

The second iteration of improvements is based on the results of the user study. This may include revisions of previous improvements and implementation of further usability features.

4 Timeline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